

# OMID ZANGANEH

## Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration  
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### PROFESSIONAL SUMMARY

Senior GIS Developer and Analyst with a track record of turning manual spatial workflows into reliable production tools. Specializes in ArcGIS Pro SDK (.NET/C#/WPF), Python geoprocessing, SQL Server pipelines, and applied AI for engineering teams. Proven results: ~90% fewer manual GIS steps, multi-day workflows reduced to minutes, and AI-assisted contract sourcing in hours instead of months. 2025 Edison Award Nominee and 2026 Edison Award Winner at Olsson. MS Geography (GIS&T), University of Nebraska at Omaha, 4.0 GPA.

### WORK EXPERIENCE

**Olsson** ♦ 2025 Nominee & 2026 Edison Award Winner

March 2025 – Present

*GIS Developer and Analyst*

Lincoln, Nebraska

- Worked directly with many telecom and infrastructure clients, delivering GIS tools, network design support, and spatial analysis that improved speed and consistency of delivery.
- Built production Python and C# GIS apps, including bore profile automation that cut processing from days to minutes.
- Created an ArcGIS Automation Suite of Python toolboxes that reduced manual GIS steps by ~90% and sped up fiber network design.
- Shipped ArcGIS Pro add-ins (.NET/C#/WPF) adopted across projects and teams — including a GIS Data Downloader used firm-wide, FTTH Network Designer, RF Analysis panel, and Street View tool.
- Learned iBwave to deliver DAS (distributed antenna system) designs, and built AI tools (Azure OpenAI, Google AI Studio, YOLO) that cut contract sourcing from months to hours.

**University of Nebraska at Omaha**

January 2024 – August 2025

*Graduate Teaching Assistant — Instructor of Record*

Omaha, Nebraska

- Taught Human-Environment Geography labs as sole instructor of record to 150+ students across three semesters.

**University of Nebraska at Omaha**

June 2024 – August 2025

*GIS Technician — Omaha Spatial Justice Project*

Omaha, Nebraska

- Digitized historical parcels and legal records into a geodatabase of racially restrictive covenants in Douglas County.

### SELECTED PROJECTS

**RFP Radar** — AI-Powered RFP Sourcing

Azure OpenAI · Google Gemini · Playwright · Python/C#

Desktop app for web-grounded RFP search, classification, and CSV export. Months of sourcing → hours.

**Bore Profile Automation** — Directional Drilling Profile Generator

Python · C# · ArcGIS Pro · SQL Server · Matplotlib

Reads spatial waypoints and elevation models; generates 2D/3D bore profiles. Days → minutes.

**Fiber Automatic Expansion** — Density Grouping & Fiber Build-Area Planning

C# · .NET · ArcGIS Pro SDK · WPF · Spatial Analysis

Grids the study area for density analysis, flood-fills viable build zones by PPM thresholds, and generates economic scorecards. Used across multiple clients; days/weeks → minutes.

**FTTH Network Designer** — Automated Fiber Optic Network Planning

C# · .NET 8 · ArcGIS Pro SDK · WPF · Kruskal MST

Places shafts, connects homes, and builds an optimal Main Trunk / Terminal Branch backbone. Days → minutes.

## ArcGIS Data Downloader Add-in — Multi-Source GIS Data Acquisition

C# · .NET 8 · ArcGIS Pro SDK · WPF · REST APIs

Downloads OSM, USGS, FEMA, Census/TIGER, Wikipedia, and BSL layers into ArcGIS Pro. Adopted firm-wide across projects and teams.

## RF Analysis Tool — 8-Tool Wireless Planning Panel

C# · ArcGIS Pro SDK · Python

Coverage prediction, PCI/RSI planner, tilt/azimuth optimizers, interference analysis, and tower placement — all as map layers.

## Aerial & Streetview AI Detection — YOLO Utility Infrastructure Detection

Python · YOLO · OpenCV · Aerial/Street Imagery APIs

Detects and classifies utility assets from aerial and street imagery; exports georeferenced results to ArcGIS.

## GeoPipe — Enterprise GIS ETL to SQL Server

Python · CustomTkinter · pyodbc · SQL Server · PyInstaller

GUI ETL for large spatial/tabular imports with schema auto-detection, GEOMETRY/GEOGRAPHY support, and connection-loss auto-resume.

## omidzanganeh.com — Full-Stack Portfolio & Browser GIS Tools

Next.js 16 · React · TypeScript · Leaflet · Vercel

Personal site with 10+ browser GIS tools (geocoder, isochrone, elevation, census, coordinate converter), Gym Flow, Stock Screener, and an AI news aggregator.

## EDUCATION

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### Master of Science, Geography — Geographic Information Science & Technology August 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

**Thesis:** Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, and environmental justice.

### Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

## TECHNICAL SKILLS

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**Languages & Frameworks** Python · C# / .NET · SQL · TypeScript · JavaScript · HTML/CSS · React · Next.js · WPF

**GIS & Spatial** ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes (.pyt) · ArcGIS Online · ArcGIS Enterprise · QGIS · Geoprocessing · Network Analysis · Remote Sensing · Google Earth Engine · iBwave

**AI / ML** Azure OpenAI · Azure AI Foundry · Google AI Studio / Gemini · YOLO · OpenCV · Prompt Engineering · Web Grounding · Batch Classification Pipelines

**Data & Backend** SQL Server · Supabase · PostgreSQL · pyodbc · ETL Pipelines · Spatial Types (GEOMETRY/GEOGRAPHY) · Smartsheet API

**Web & Cloud** Next.js App Router · Vite · Leaflet · Vercel · REST APIs · PWA · Microsoft Azure · Google Cloud

**Desktop & Tooling** CustomTkinter · PyInstaller · Playwright · Matplotlib · AutoCAD · ENVI · SNAP · Photomod · Tableau · Adobe Photoshop · Adobe Illustrator

**Languages** English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: [omidzanganeh.com](https://omidzanganeh.com)

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